

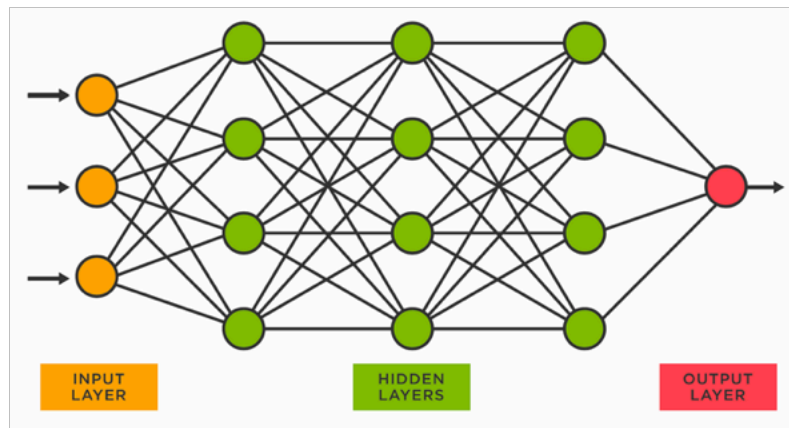
The problem of interest:

$$\min_x f(x), \quad f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad f(x) = \sum_{i=1}^m f_i(x)$$

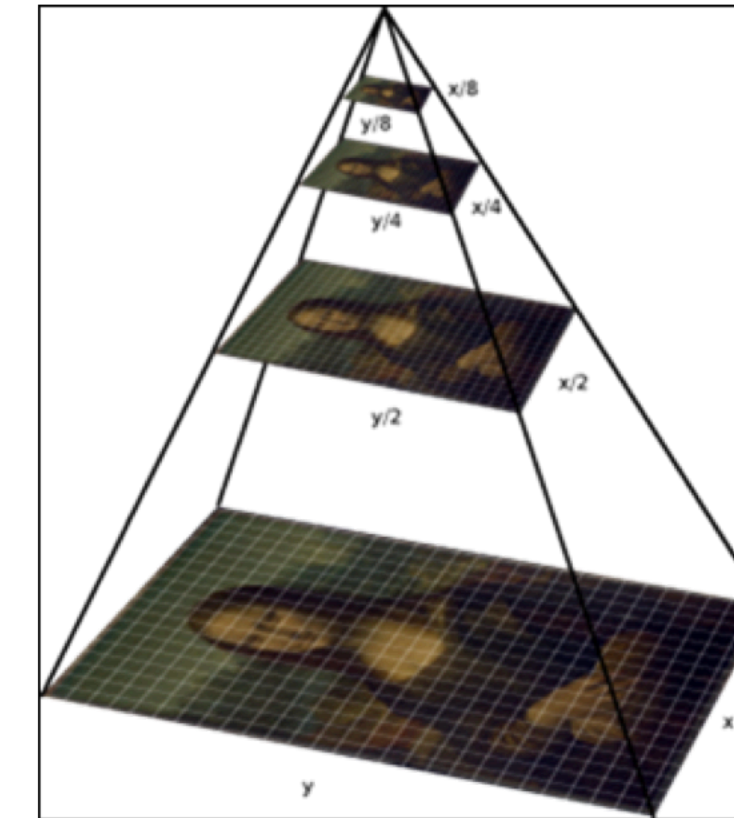
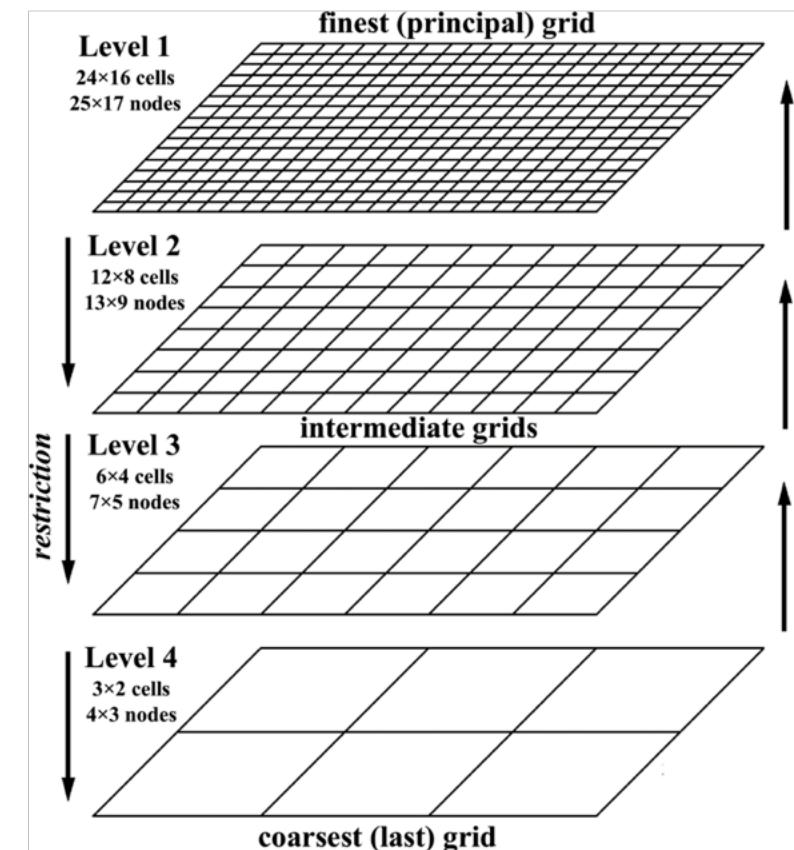
f_i non-linear, non-convex, n, m very large

Examples:

- Image restoration
- Sparse matrix factorization
- Neural networks training



A special class of problems: hierarchically structured problems



Objectives:

Optimization problem

High-dimensional
data

Complex models

Exploiting
hierarchical
structures

Parsimonious and
mathematically founded
optimization methods